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Enhanced hydrolysis and methane yield by applying microaeration pretreatment to the anaerobic co-digestion of brown water and food waste

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ABSTRACT

Microaeration has been used conventionally for the desulphurization of biogas, and recently it was shown to be an alternative pretreatment to enhance hydrolysis of the anaerobic digestion (AD) process. Previous studies on microaeration pretreatment were limited to the study of substrates with complex organic matter, while little has been reported on its effect on substrates with higher biodegradability such as brown water and food waste. Due to the lack of consistent microaeration intensities, previous studies were not comparable and thus inconclusive in proving the effectiveness of microaeration to the overall AD process. In this study, the role of microaeration pretreatment in the anaerobic co-digestion of brown water and food waste was evaluated in batch-tests. After a 4-day pretreatment with 37.5 mL-O_2/L_R-d added to the liquid phase of the reactor, the methane production of substrates were monitored in anaerobic conditions over the next 40 days. The added oxygen was consumed fully by facultative microorganisms and a reducing environment for organic matter degradation was maintained. Other than higher COD solubilization, microaeration pretreatment led to greater VFA accumulation and the conversion of other short chain fatty acids to acetate. This could be due to enhanced activities of hydrolytic and acidogenic bacteria and the degradation of slowly biodegradable compounds under microaerobic conditions. This study also found that the nature of inoculum influenced the effects of microaeration as a 21% and 10% increase in methane yield was observed when pretreatment was applied to inoculated substrates, and substrates without inoculum, respectively.

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1. Introduction

Anaerobic digestion (AD) is a four step biological process known to be effective in converting organic waste into methane which is a potentially valuable by-product that can be used as a fuel in producing heat and electricity. This is in line with the shifting emphasis from using traditional fossil fuels to the use of renewable energy. The AD process is thus a sustainable environmental technology that addresses the global concerns over pollution control and energy shortages.

The first two steps of the AD process – hydrolysis and acidification involve the solubilization of complex particulate organic compounds into simple soluble compounds such as volatile fatty acids (VFAs). They are followed by the acetogenesis step which converts the VFAs to acetate and hydrogen gas that would in turn be consumed by methanogens to produce methane in the final step of the AD process (Batstone et al., 2002). As hydrolysis is usually the rate-limiting step for substrates in particulate form, pretreatments that enhance hydrolysis efficiencies are commonly employed to improve the overall AD process. In order for AD systems to be applicable to land scarce urban cities, there is a need for compact anaerobic digesters. This could be partly achieved by shortening the hydraulic retention time of the AD process by employing suitable pretreatment methods.

Several methods such as mechanical, ultrasonic, thermal, chemical, and biological pretreatments were shown to improve methane production in anaerobic digesters (Dhar et al., 2012; Vavouraki et al., 2012; Benabdallah El-Hadj et al., 2007). However, pretreatments often lead to increased capital costs due to the additional energy or chemicals required (Bordeleau and Droste, 2011). In some cases, the additional methane produced due to pretreatments was insufficient to recover the additional costs (Tartakovsky et al., 2011; Liu et al., 2008) thus rendering the process not feasible financially.

The term "microaeration" was defined by Botheju and Bakke (2011) as the introduction of small amounts of oxygen into an anaerobic biochemical process to enable both anaerobic and aerobic biological activities to occur within a single bioreactor.

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Microaeration has been used conventionally for the desulphurization of biogas (Fdz-Polanco et al., 2009), and recently it was shown to be an alternative pretreatment to enhance hydrolysis of complex organic matter.

Although oxygen is known to induce inhibitory effects on methanogens which are strict anaerobes, advantages of limited aeration in AD systems have been reported. Earlier studies have shown that limited aeration could reduce formation of toxic metabolites such as lactic acid and ethanol (Zeng and Deckwer, 1996) as well as promote the synthesis of certain lipids required for the stability of anaerobes cell membrane (Ghaly and El-Taweel, 1995). Limited aeration was also shown to enhance the hydrolysis of carbohydrates and proteins (Johansen and Bakke, 2006) as well as the activities of cellulase and protease hydrolytic enzymes (Charles et al., 2009). Recent studies that discussed the effects of microaeration on AD processes are summarized in Table 1. Some of them reported a positive impact on hydrolysis efficiency (Johansen and Bakke, 2006; Zhu et al., 2009; Jagadabhi et al., 2010) and a shorter lag-phase time achieved with microaeration (Díaz et al., 2011). Díaz et al. (2011) found that limited oxygen supply to anaerobic digesters did not inhibit methane production and other studies reported that microaeration improved methanogenic activity (Zitomer and Shrout, 1998) and methane yield (Tartakovsky et al., 2011; Botheju et al., 2010; Nguyen et al., 2007). As shown in Table 1, Botheju et al. (2010) reported a 30–55% increase in methane yield due to improved solubilization of starch, while another study conducted by Tartakovsky et al. (2011) showed a 26% increase in methane yield when oxygen produced from the electrolysis of water was applied to the anaerobic digestion of a concentrated stock solution of synthetic wastewater. The enhanced methane production was due to both increased hydrolysis and the conversion of electrolytic hydrogen to methane by hydrogenotrophic methanogens.

However, enhanced hydrolysis under high oxygen levels did not translate to higher methane potential for studies by Zhu et al. (2009) and Johansen and Bakke (2006). In another study, limited aeration brought about a 20% higher methane production but the washout of methanogens and accumulation of substrate were observed as oxygen flux was further increased (Gerritse et al., 1990). Excessive aeration was found to decrease the methane potential of AD systems due to higher aerobic respiration and lower anaerobic fermentation leading to increased biomass and CO_2 generation but lower VFAs and soluble COD levels. In addition, facultative organisms have higher growth rates and would out-compete strict anaerobes under high oxygen levels due to substrate competition. Although the introduction of oxygen could influence a range of reactions in AD systems, Joss et al. (1999) reported that microaeration in AD systems did not affect other biochemical interactions significantly other than aerobic respiration and carbon dioxide generation.

The increasing amount of food waste generated worldwide is a growing concern. According to US EPA (2002), approximately 43.6 million tons of food was thrown away each year in the United States. Another study revealed that British households generate approximately 1.8 million tons of food waste yearly (WRAP, 2009) while in Singapore, the annual food waste generation in 2011 was 605,800 tons with only 10% being recycled (Singapore waste statistics, 2011). In most countries, food waste is mostly disposed in landfill but the high organic content of food waste leads to uncontrolled emission of landfill gas which poses serious environmental problems. In view of rising costs for waste disposal as well as depleting energy resources, the AD of food waste was found to be a more sustainable treatment method due to the high degree of waste stabilization and methane generation (Wang et al., 2005).

Studies on the co-digestion of black water (wastewater containing feces and urine, collected from vacuum toilets) and food waste showed positive synergistic effects of co-digestion in the form of increased methane production and COD removal rates (Wendland et al., 2007; Kujawa-Roeleveld and Zeeman, 2006). When food waste was added to brown water (wastewater containing feces only) instead of black water, both methanization and methane production further increased from 34% to 61% and from 26.5 to 50.8 L CH_4/p/d, respectively (Kujawa-Roeleveld et al., 2003). Other than being more diluted, black water also contains high amounts of ammonia, due to the presence of urine, which might inhibit the AD process. Hence, the co-digestion of brown water and food waste would be preferred for more efficient AD processes.

Previous studies on the effects of microaeration in AD systems showed that it led to improved hydrolysis for substrates with complex organic matter, such as sludge, starch, grass-silage and cellulose. However, the effects of microaeration pretreatment on the hydrolysis of substrates with higher biodegradability, such as brown water and food waste, have yet to be reported. Due to the lack of consistent microaeration intensities, previous studies were not comparable and thus inconclusive in proving the effectiveness of microaeration to the overall AD process. It is also unclear if microaeration will impact biochemical processes other than hydrolysis in the AD process. The aim of this study was to investigate the effect of microaeration pretreatment on the degradation of brown water and food waste in batch tests. The effects of

Table 1
Summary of literature carried out on the effects of microaeration on the anaerobic digestion process at mesophilic conditions (35–37 °C).

Material	Reactor scale	Reactor volume (L)	Aeration intensity	Effect on hydrolysis efficiency	Effect on degree of acidogenesis	Effect on methane yield	Reference
Primary sludge	Batch (4 days)	0.5	33.3 L air/kg TS/d	50–60% increase	–	Negative impact	Johansen and Bakke (2006)
Vegetable and flower waste	Batch (5 days)	1.4	74, 147, 442, 1768 L air/kg TS/d	Positive impact	Negative impact	–	Zhu et al. (2009)
Starch	Batch (17 days)	0.1	16% of feed COD. Air added once at beginning	–	–	30–55% Increase	Botheju et al. (2010)
Grass-silage	Batch, Leach Bed Reactor (60 days)	1.0	2.5 L air (1 L/min)	Positive impact	4-fold increase	–	Jagadabhi et al. (2010)
Cellulose	Batch (19 days)	2.1	0.01 L O_2/d	Positive impact (shorter lag-phase time)	–	No impact	Díaz et al. (2011)
Synthetic wastewater	Batch UASB Reactor, HRT = 6 h	0.5	3.24×10^{-4} mL O_2/min, from the electrolysis of water	–	–	26% Increase	Tartakovsky et al. (2011)
Brown water and food waste	Batch (45 days)	0.25	0.0375 L $O_2/L_{reactor}$/d	Positive impact	Positive impact	10–21% Increase	This Study

microaeration on the hydrolysis, acetogenesis and methanogenesis steps would be studied by comparing the hydrolysate from reactors pretreated with microaeration and the hydrolysate from the anaerobic control reactors during the 4-day pretreatment. This would be followed by the monitoring of methane production from both reactors over the next 40 days. The influence of inoculum on the effects of microaeration would also be investigated.

2. Materials and methods

2.1. Substrate and inoculum

Food waste was collected from one of the canteens at Nanyang Technological University campus, where the majority of the waste came from Chinese, Indian, Indonesian and Malay food stalls. It was a mixture of cooked meat, rice, noodles, vegetables and salad. After removing bones and non-food materials, the food waste was blended into slurry form with an average size of 2 mm using a food blender. The blended food waste was then mixed well, and stored in a refrigerator at 4 °C. Brown water was collected from a specially designed source-separation toilet located in the laboratory, where urine with 0.3-L flush water (yellow water) and feces with 2.0-L flush water (brown water) was collected in separate tanks. The brown water collected was stored in a refrigerator at 4 °C. The substrate for this batch study consisted of a mixture of food waste and brown water in a ratio of 150-g:2-L. Mesophilic anaerobic sludge used in this study was collected from a local wastewater treatment plant (Ulu Pandan Water Reclamation Plant, Singapore). Before use, the sludge was stored at 35 °C for 1 week to remove any remaining biodegradable fraction of the sludge and was sieved through a 1 mm mesh to remove any large particulates.

2.2. Batch reactors

Batch reactors were operated using 0.25 L continuously stirred reactors with a working volume of 0.2 L. Two series of experiments were conducted in parallel. The first one consisted of the substrate seeded with sludge in a ratio of 50:50 (v/v), giving rise to a $VS_{inoculum}/VS_{substrate}$ ratio of 0.5. The other series consisted of only the substrate, and was seeded with sludge after the 4-day pretreatment. The contents in the digesters were flushed with N_2 to provide anaerobic conditions and incubated at 35 °C with mixing at 100 rpm. The digesters were fed once at the beginning with the biogas being monitored, and hydrolysate/digestate extracted regularly using a syringe. The microaeration pretreatment was applied from the start for four consecutive days where 37.5 mL O_2/L_R-day was added to the liquid once daily. Due to the relatively low $VS_{inoculum}/VS_{substrate}$ ratio of 0.5, biogas production was observed over 40 days in this study. Biogas production as well as the quality of the digestate were monitored throughout the 40 days of batch study. The performance of the digesters pretreated with microaeration was compared to a control with no pretreatment. All experiments were conducted in duplicates.

2.3. Analytical procedures

Biogas production and its composition were measured once every 2–4 days for the first 15 days and subsequently once every 5–8 days for the rest of the batch study to allow accumulation of biogas sufficient for the analysis of biogas composition using the GC–TCD. The extracted hydrolysate/digestate was measured for its pH, oxidation–reduction potential (ORP), VFA and chemical oxygen demand (COD). pH, ORP, VFA and COD levels were measured daily for the initial 4 days when microaeration pretreatment was applied. pH, VFA and COD levels were subsequently measured once weekly until the end of the batch study. Total solid (TS) and volatile solid (VS) were measured only at the beginning and end of the study. pH value was measured using a compact titrator (Mettler Toledo) equipped with a pH probe (Mettler Toledo DGi 115-SC). ORP was measured by a platinum ORP combination electrode (Fisher Scientific Accumet® Ca^+ Model 13-620-81). TS and VS were analyzed according to the Standard Methods (APHA, 1998). Total and soluble chemical oxygen demand were measured using COD digestion vials (Hach Chemical) and a spectrophotometer (DR/2800, Hach). Soluble COD was measured using the supernatant of samples after centrifugation (KUBOTA 3700, Japan) at 10,000 rpm for 15 min. The determination of VFAs was carried out using a gas chromatograph (Agilent Technologies 7890A, USA), equipped with a flame ionization detector (FID) and a DB-FFAP (Agilent Technologies, USA) column (30 m × 0.32 mm × 0.50 μm) and the samples were filtered through Membrane Solutions 0.45 μm cellulose acetate membrane filters. Total biogas production was monitored using a syringe, while the biogas composition (methane, carbon dioxide, nitrogen and hydrogen contents) was analyzed by gas chromatograph (Agilent Technologies 7890 A, USA) equipped with a thermal conductivity detector (TCD).

2.4. Calculation

Methane production was determined by subtracting the methane produced by inoculum from that produced in the reactors containing substrate and inoculum. Cumulative methane yield was calculated as the sum of methane produced throughout the study period and expressed as liters per kilogram of VS of substrate fed to the reactors.

3. Results and discussion

The effect of applying microaeration pretreatment to the co-digestion of brown water and food waste was evaluated in batch reactors. The 4-day microaeration pretreatment was applied to reactors (I) containing substrate inoculated with sludge, and (II) containing substrate only. Sludge was added to reactor (II) after the 4-day pretreatment, and methane production from both reactors were monitored over the next 40 days under completely anaerobic conditions. The hydrolysate and methane yields of reactors with 4-day microaeration pretreatment (MA) were compared to a control that was operated under fully anaerobic conditions throughout the study (AN).

3.1. Reactor (I): pretreatment applied to substrate inoculated with sludge

The soluble COD levels of hydrolysate from MA were 6.62 ± 0.01, 7.00 ± 0.08, 8.24 ± 0.06 and 6.88 ± 0.06 g/L, while those from AN were 6.00 ± 1.00, 5.42 ± 0.08, 5.68 ± 0.03 and 3.06 ± 0.04 g/L after 1-, 2-, 3- and 4-day pretreatment, respectively. This corresponded to 9%, 23%, 31% and 56% higher solubilization for MA after 1-, 2-, 3- and 4-day pretreatment, respectively. The ORP was used to monitor the level of oxygen in the reactors since it could sense the dosing of oxygen even at a level well beyond the detection limit of commercially available oxygen probe. ORP varies between −200 and −300 mV in fully anaerobic systems, and increases to above 50 mV in aerobic systems (Jenicek et al., 2008). The ORP of MA ranged between −220 and −278 mV throughout the 4-day microaeration pretreatment. This was similar to the ORP of AN (−253 to −270 mV), indicating that the oxygen added to MA (37.5 mL-O_2/L_R-d) was consumed by the facultative microorganisms and both reactors had anaerobic zones where fermentation could occur.

In this study, short-chain fatty acids (SCFAs) i.e. acetate (C2), propionate (C3), butyrate (C4), iso-butyrate (i-C4), valerate (C5), iso-valerate (i-C5), caproate (C6), iso-caproate (i-C6) and heptanoate (C7) were analyzed daily during the 4-day pretreatment. The concentrations of SCFAs in the hydrolysate after 1-, 2-, 3- and 4-day pretreatment are shown in Fig. 1a. The total VFA production for MA was 2.32 ± 0.07, 5.90 ± 1.28, 7.06 ± 0.37 and 5.57 ± 0.07 g-COD/L after 1-, 2-, 3- and 4-day pretreatment, respectively. This was 2%, 76%, 35% and 330% higher than that of AN on the respective days of pretreatment. The composition of each SCFA was calculated by taking the ratio of individual SCFA concentration over that of TVFA. The compositions of acetate and propionate in the hydrolysate of MA were on average 10% higher and 12% lower, respectively than that of AN during the 4-day pretreatment, while the compositions of the remaining SCFAs in the hydrolysate of MA and AN were not significantly different. This suggested that microaeration enhanced the breakdown or oxidation of propionate to acetate. Propionate degrading organisms are known to be the slowest growing and the most sensitive acetogenic organisms present in most anaerobic digester ecosystems (Wall et al., 2008). Thus, the accumulation of propionate in reactors often leads to inhibition and reduced organic removals. On the other hand, acetate is considered to be a direct substrate for methanogens. The reduced levels of propionate and increased production of acetate after pretreatment would be beneficial for subsequent methanogenesis.

The cumulative biogas yields of MA and AN were 530 ± 36 and 431 ± 13 L/kg VS_{fed}, respectively (Fig. 2a). The composition of biogas was tested for hydrogen, methane and carbon dioxide contents. Hydrogen was present in either small amounts (less than 2% of biogas volume) or not detected in both MA and AN in this study. As shown in Fig. 3a, the methane yield of the pretreated substrate (MA), after the first day of pretreatment was 14 ± 1 L/kg VS_{fed}, 34% higher than that of the control (AN). However, the methane yields for both MA and AN were similar for the subsequent 3 days. Due to the pH drop (Fig. 4a) caused by high degree of COD solubilization and VFA accumulation, the methane yield of MA was inhibited and thus lower than that of AN for the next 10 days. The improvement in methane yield due to enhanced solubilization and VFA accumulation after microaeration pretreatment was

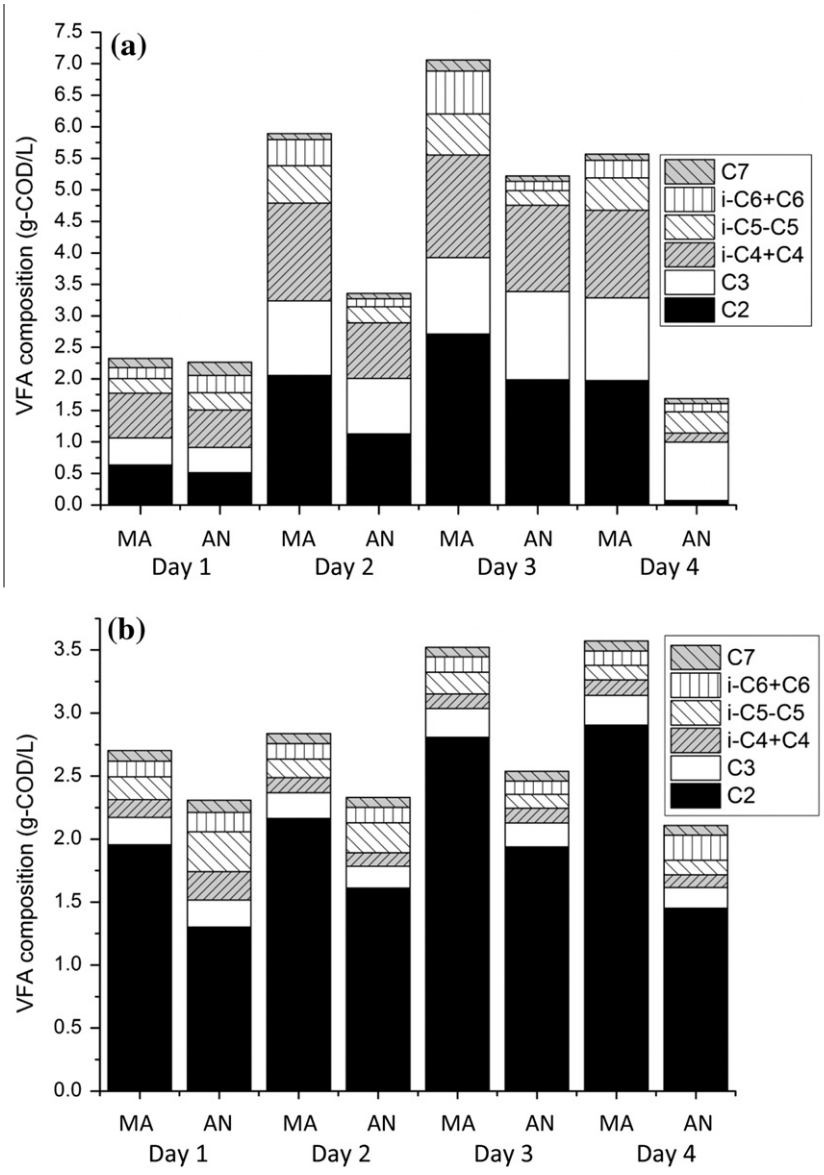


Fig. 1. Composition of volatile fatty acids for hydrolysate of: (a) reactor I, and (b) reactor II during 4-day pretreatment.

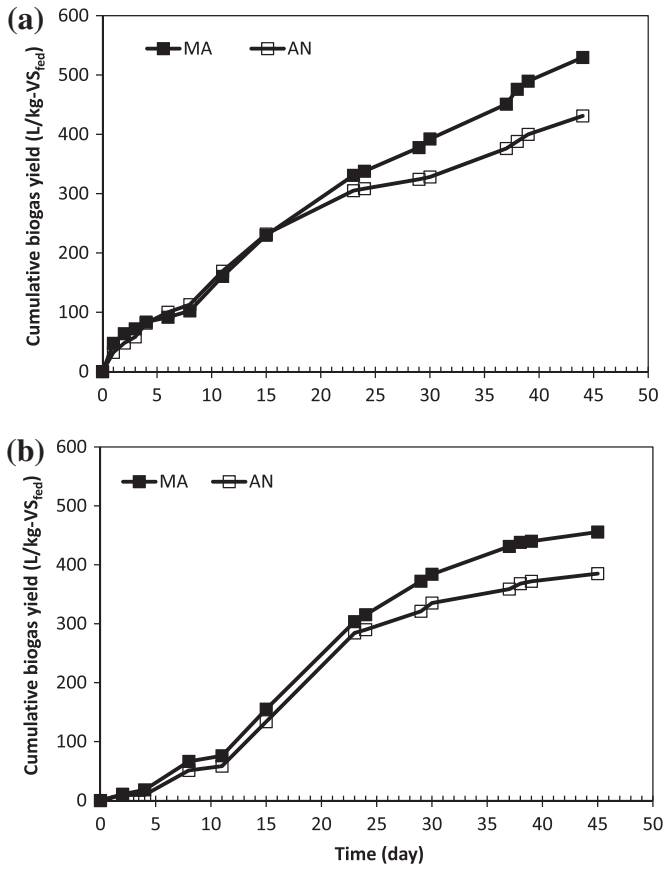


Fig. 2. Average cumulative biogas yield of –■– MA and –□– AN for: (a) reactor I, and (b) reactor II throughout the entire experiment duration.

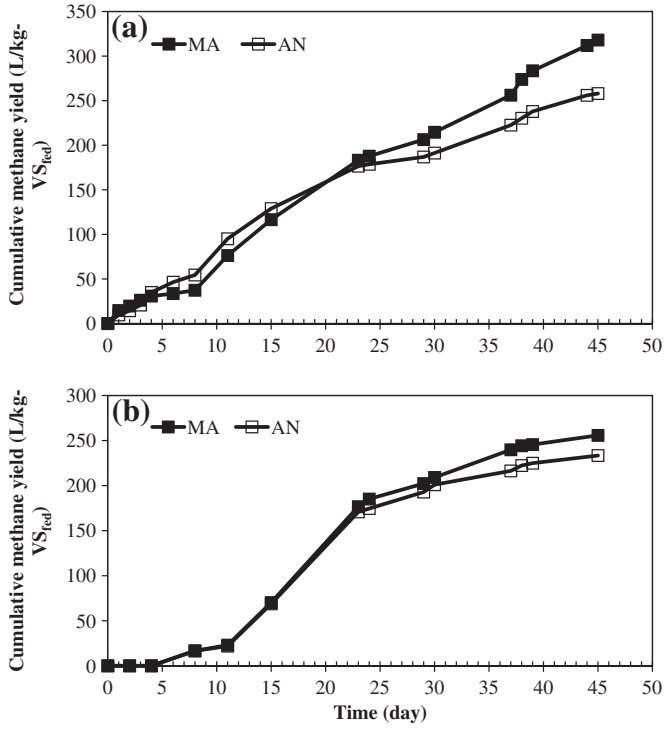


Fig. 3. Average cumulative methane yield of –■– MA and –□– AN for: (a) reactor I, and (b) reactor II throughout the entire experiment duration.

observed after day 25. At the end of this study, the cumulative methane yield for the pretreated substrate was 318 ± 8 L/kg VS_{fed}, while that of the control was 258 ± 15 L/kg VS_{fed}. Similar to the higher methane yield of MA observed after the first day of pretreatment in this study, Díaz et al. (2011) also found a shorter lag phase with the microaerobic assays for the anaerobic digestion of cellulose. The accelerated solubilization of cellulose in the very early stages of the digestion led to higher methane production during the first few days. However, the final methane production of cellulose after 19 days was not significantly different between the anaerobic and microaerobic assays. In contrast to the findings of Díaz, the final methane yield in this study was 21% higher for the pretreated substrates due to the higher degree of solubilization and acidification.

3.2. Reactor (II): pretreatment applied to substrate not inoculated with sludge

The soluble COD levels for the hydrolysate in MA were 8.02 ± 0.31, 8.50 ± 0.00, 8.01 ± 0.04 and 8.76 ± 0.06 g/L while that for AN were 13.50 ± 0.25, 14.20 ± 1.13, 13.63 ± 0.04 and 13.79 ± 0.07 g/L after 1-, 2-, 3- and 4-day pretreatment, respectively. The ORP of MA ranged between −17 and −67 mV throughout the 4-day microaeration pretreatment. This was similar to the ORP of AN (−20 to −64 mV), indicating that the oxygen added to MA (37.5 mL-O_2/L_R-d) was consumed by the facultative microorganisms and both reactors had anaerobic zones where fermentation could occur.

The concentrations of SCFAs in the hydrolysate after 1-, 2-, 3- and 4-day pretreatment are shown in Fig. 1b. The total VFA production for MA was 2.70 ± 0.22, 2.84 ± 0.12, 3.52 ± 0.55 and 3.57 ± 0.48 g-COD/L after 1-, 2-, 3- and 4-day pretreatment, respectively. This was 17, 22, 39 and 78% higher than that of AN on the respective days of pretreatment. The composition of acetate in the hydrolysate of MA was on average 10% higher than that of AN during the 4-day pretreatment. Similar to the observations for reactor (I), the application of microaeration pretreatment enhanced the conversion of other SCFAs to acetate for reactor (II). It was also observed that the extent of VS degradation was 10% higher for MA than for AN, suggesting that the application of microaerobic conditions could have led to the production of exoenzymes by aerobic microorganisms, which degraded compounds that were otherwise resistant to degradation under fully anaerobic conditions. In an earlier study, Hasegawa and Katsura (1999) also found that microaerobic conditions stimulated the production of exoenzymes by aerobic microorganisms.

As shown in Fig. 4b, the pH of pretreated substrate (3.0–3.5) was slightly lower than that of the control during pretreatment, reflecting the higher VFA accumulation in MA. After 4 days, the substrate was seeded with sludge and a rise in pH to levels above 6.5 was observed. Subsequently, the pH of both MA and AN were not significantly different.

Fig. 2b shows the cumulative biogas yields and composition of biogas for MA and AN. The cumulative biogas yield of MA and AN were 456 ± 23 and 385 ± 9 L/kg VS_{fed}, respectively. Hydrogen was present in negligible amounts (less than 2% of biogas volume) while the average methane and carbon dioxide contents for MA was 64% and 36%, and that for AN was 62% and 38%, respectively.

Fig. 3b shows that the methane yields of MA and AN were not significantly different for the first 24 days, at 185 ± 7 and 175 ± 3 L/kg VS_{fed}, respectively. However, the final cumulative methane yield of MA was 256 ± 9 L/kg VS_{fed}, while that for AN was 233 ± 17 L/kg VS_{fed}. The improvement of 10% in final methane yield after microaeration pretreatment could be due to the degradation of slowly biodegradable compounds which were otherwise resistant to degradation under fully anaerobic conditions.

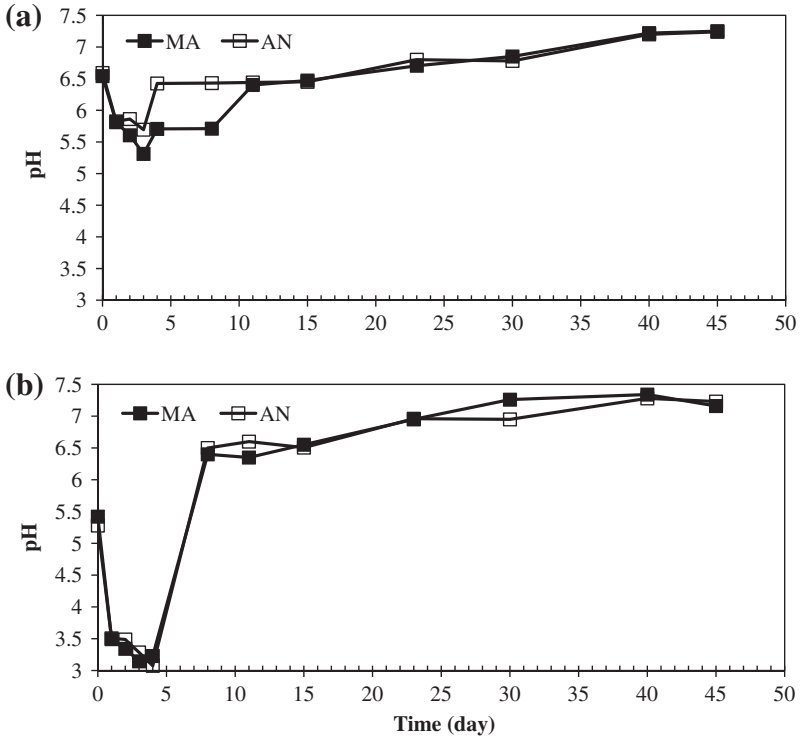


Fig. 4. Average pH levels of –■– MA and –□– AN for: (a) reactor I, and (b) reactor II throughout the entire experiment duration.

3.3. Comparison between reactors (I) and (II)

As described in Sections 3.1 and 3.2, the enhancements of COD solubilization and acidification after pretreatment were more significant for the substrate when inoculated with sludge. As a result, methane yield was improved by 21% when pretreatment was applied to the inoculated substrate, while the improvement in methane yield was 10% when pretreatment was applied to the substrate only. The application of microaeration to the inoculum could have enhanced the production and activities of hydrolytic and acidogenic bacteria present in it, giving rise to higher hydrolysis and acidification efficiencies. Botheju et al. (2010) also found that the nature of inoculum played a significant role on the level of oxygen induced enhancement of the methane yield. In contrary to the negative perception of the inhibitory and toxic effects of oxygen on methanogens, the application of microaerobic conditions to the inoculum did not cause any inhibition effects to the AD process.

In both reactors, the composition of acetate in the hydrolysate was 10% higher after microaeration pretreatment than in the control, suggesting the improved conversion of other SCFAs to acetate. Another observation from both reactors was the pH drop induced by microaeration pretreatment. As shown in Fig. 4, pH levels of MA were in general lower than that of AN during the 4-day microaeration pretreatment period due to enhanced COD solubilisation and VFA accumulation. The lower pH of MA also led to lower methane yields, as compared to AN, for the initial 15 days of batch study (Fig. 3a). Hence, another potential advantage of applying microaeration pretreatment would be to prevent methane formation in the pretreatment stage of AD.

3.4. Final discussion

Hydrolysis is the breakdown and solubilization of particulate matter by extracellular enzymes excreted by hydrolytic microorganisms. Because of the higher hydrolytic enzyme production, hydrolysis efficiency is known to be higher under aerobic as compared to anaerobic conditions (Johansen and Bakke, 2006; Gujer et al., 1995). The intensity of microaeration determines both the hydrolysis efficiency and the degree of acidification, thus affecting the methane yield of the AD process. At a microaeration intensity of 33.3 L air/kg TS/d, Johansen and Bakke (2006) found a 50–60% increase in hydrolysis efficiency but a negative impact on methane yield during the anaerobic digestion of primary sludge. Similarly, Zhu et al. (2009) found a positive impact on hydrolysis efficiency but a negative impact on acidogenesis when a microaeration intensity of 74 L air/kg TS/d was applied to the anaerobic digestion of vegetable and flower waste. At these microaeration intensities, the extra solubilized organic matter was converted into CO_2 and new biomass by the aerobic respiration of facultative microorganisms, leaving less substrate for methanogenesis. To ensure the conversion of extra solubilized organic matter into methane, an increase in the degree of COD solubilization and the accumulation of VFAs must be achieved.

At a microaeration intensity of 37.5 mL-O_2/L_R-d, enhanced solubilization of organic matter without the oxygen-related inhibition of acidogenesis was achieved in this study. The accumulation of VFA achieved with microaeration pretreatment was also observed previously by Hasegawa et al. (2000), who found that the thermophilic pretreatment of sludge under microaerobic conditions led to VFA accumulation, and Jagadabhi et al. (2010), who found a 4-fold increase in VFA levels upon inducing microaerobic conditions (2.5 L air at 1 L/min) to the fermentation of grass-silage.

In this study, the enhanced solubilization and acidification achieved with microaeration pretreatment at 37.5 mL-O_2/L_R-d resulted in 21% and 10% higher methane production for the anaerobic co-digestion of brown water and food waste in reactors (I) and (II), respectively. However, the improvement in methane yield by microaeration pretreatment was only observed after day 25. This was due to the inhibition of methanogenesis at low pH levels caused by enhanced hydrolysis and acidification in MA. The enhancement of methane yield by microaeration pretreatment could be observed earlier if the $VS_{inoculum}/VS_{substrate}$ ratio was higher.

Unlike other pretreatment methods that improve the overall AD process by increasing the hydrolysis efficiency, this study showed that microaeration pretreatment could also enhance the conversion of other SCFAs to acetic acid, and no inhibition of methanogenesis was observed. Microaeration pretreatment does not require the addition of chemicals but only a limited supply of air. Thus, it could be a cost effective pretreatment since anaerobic digesters are inevitably subjected to varying levels of limited and unintended aeration due to activities such as feeding and effluent removal. As pointed out by Botheju et al. (2010), the optimum oxygenation level for maximum methane yield of an AD system could depend on various factors such as the reactor biomass concentration and feed composition. Hence, one oxygen supply rate that worked well in one AD system may not do so in others. We could thus rely on ORP readings to control the amount of oxygen entering the AD systems. ORP sensors could be integrated in anaerobic digesters to monitor or control the intensity of applied microaeration, so as to avoid aerobic respiration and damage to those critical strict anaerobic microorganisms due to excess oxygen. However, further investigations are still required to develop a suitable operating system for the application of microaeration pretreatment in AD systems.

4. Conclusion

This study demonstrated the benefits of adding small amounts of oxygen in AD systems in batch test. The results concluded that microaeration is an effective pretreatment for the anaerobic co-digestion of brown water and food waste. Besides increased solubilization and acidification efficiencies, microaeration enhanced the breakdown of other SCFAs to acetic acid. The application of microaeration pretreatment was also suggested to stimulate the production of exoenzymes that carried out the degradation of slowly biodegradable compounds which were otherwise resistant to degradation under fully anaerobic conditions. The more significant enhancement in solubilization and acidification efficiencies observed for the inoculated substrate after pretreatment suggested that the nature of inoculum influenced the effects of microaeration, likely through the higher production and activities of hydrolytic and acidogenic bacteria present in the inoculum under microaerobic conditions. The microaeration pretreatment was sufficient to increase the degree of solubilization but not enough for the subsequent VFA oxidation, thus resulting in higher methane yields. The accumulated VFA achieved with microaeration pretreatment resulted in 21% and 10% higher methane yield when pretreatment was applied to inoculated substrates, and substrates without inoculum, respectively.

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